

Project Design Phase

Solution Architecture

Date	30 June 2026
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Project Name	Meeting Room Booking System using ServiceNow App Engine Studio
Maximum Marks	4 Marks

1. Overview

The solution architecture of the Meeting Room Booking System describes how the application is structured on the ServiceNow platform and how data flows between its components – from the moment an employee submits a booking request to the point an administrator approves it and reports are generated. The architecture is organized into five logical layers: the **User Layer**, the **Presentation Layer**, the **Application / Automation Layer**, the **Data Layer**, and the **Notification & Analytics Layer**. Each layer has a clearly defined responsibility, which keeps the system modular, easy to maintain, and simple to extend.

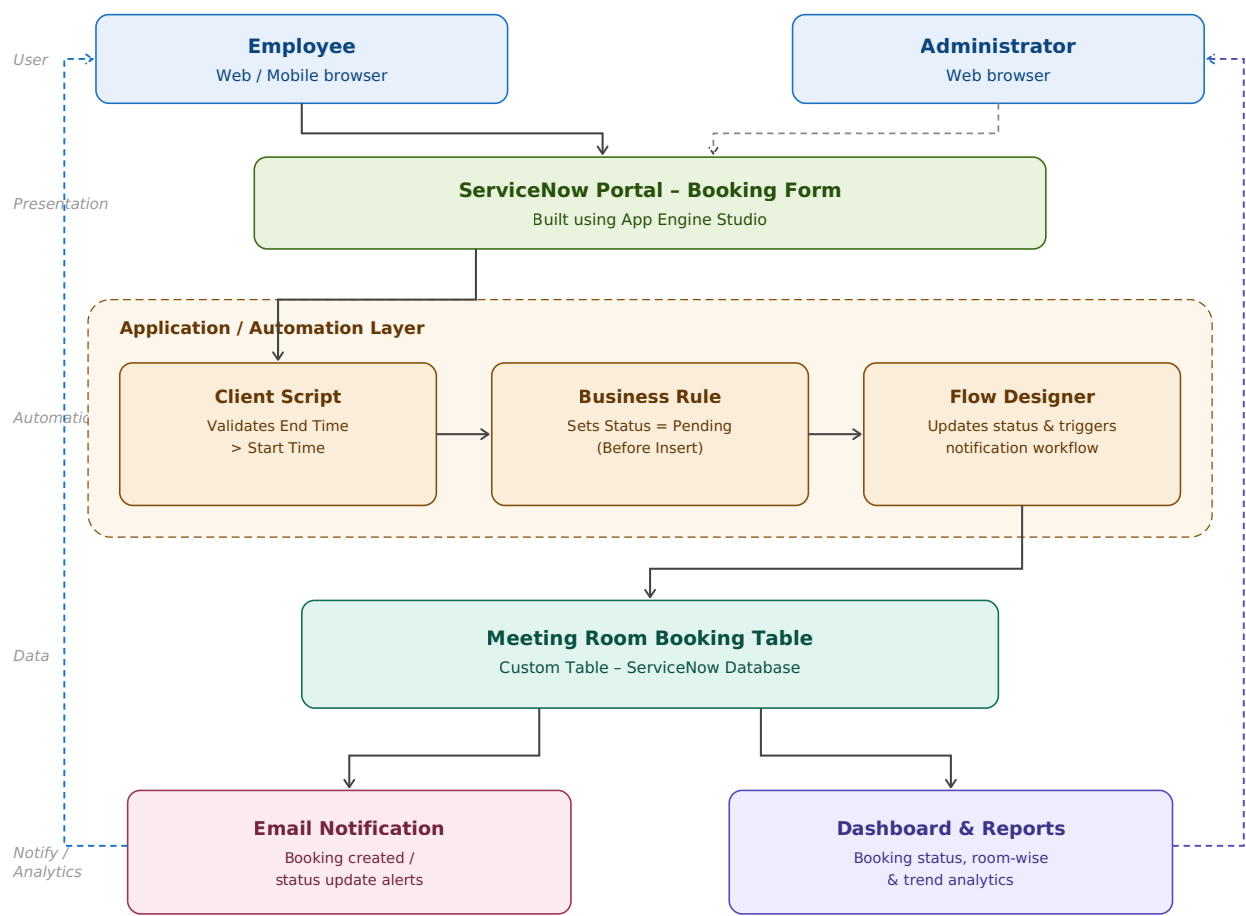


Figure 1: Solution architecture and data flow of the Meeting Room Booking System (ServiceNow App Engine Studio)

Solid arrows = primary booking flow | Dashed arrows = return/feedback flow (email alert to employee, admin review & approval)

2. Architecture Explanation

Layer	Components	Role in the Architecture
User Layer	Employee, Administrator (web / mobile browser)	Represents the two human actors who interact with the system. Employees initiate booking requests; administrators review, approve, and monitor them. Both access the application through a standard browser – no separate client installation is required.
Presentation Layer	ServiceNow Portal – Booking Form (App Engine Studio)	The booking form, built using App Engine Studio, is the single entry point for capturing employee name, employee ID, room, date, start/end time, and purpose. It presents real-time feedback to the user and forwards submitted data to the automation layer.
Application / Automation Layer	Client Script, Business Rule, Flow Designer	This is the processing core of the system. The Client Script validates that End Time is later than Start Time before the form can be submitted. The Business Rule runs on the server before insert and automatically sets the booking status to "Pending". Flow Designer then orchestrates the rest of the workflow – updating status and triggering downstream notifications – without any manual coding.
Data Layer	Meeting Room Booking Table (custom ServiceNow table)	A dedicated custom table stores every booking record with an auto-numbered ID (prefix MRB), employee details, room, schedule, purpose, and status. It acts as the single source of truth that all other layers read from and write to.
Notification & Analytics Layer	Email Notification, Dashboard & Reports	Once a record is created or updated, an automated email notification informs the employee of the booking status. In parallel, a dashboard built on the same table gives administrators live visibility into total bookings, status breakdown, room-wise usage, and booking trends, supported by dedicated reports for deeper analysis.

3. Data Flow Summary

- The employee submits the booking form through the ServiceNow portal.
- The Client Script validates the entered time range on the client side before submission.
- On submission, the Business Rule sets the initial status to "Pending" and the record is inserted into the Meeting Room Booking table.
- Flow Designer is triggered by the new record, updates the workflow status, and fires the Email Notification.
- The employee receives an email confirming that the request has been logged.
- The administrator reviews pending requests on the dashboard, then approves or completes the booking, updating the same table.
- Reports and dashboard widgets continuously reflect the latest data for monitoring and decision-making.

Reference: ServiceNow App Engine Studio Documentation – docs.servicenow.com